

KYLE NAKAMURA

Orange County, California

Phone: (626) 388-5416 | Email: knakamura13dev@gmail.com

LinkedIn: [linkedin.com/in/kylenakamura](https://www.linkedin.com/in/kylenakamura) | GitHub: github.com/knakamura13

CAREER PROFILE

Results-driven Software Engineer with 9+ years of experience and an M.S. in Computer Science specializing in Machine Learning. Currently building AI-powered higher education products at HighPoint, where I architect LLM-integrated features, design MCP server infrastructure, and lead AI developer enablement across the engineering team. Proven track record launching greenfield projects, delivering complex React/TypeScript applications, and bridging the gap between ML research and production software.

KEY SKILLS

Professional: Technical Leadership, AI Strategy & Enablement, Project Management, Agile, Mentorship

Software Dev.: Full-Stack Development, React, Angular, SvelteKit, TypeScript/JavaScript, Kotlin, PHP, GraphQL, HTML, SCSS, TurboRepo, Storybook, Playwright E2E Testing, Rapid Prototyping, CI/CD, Docker, Git/GitLab, Linux/Unix, MySQL, RESTful APIs, Cloud Services (AWS, GCP)

ML & Technical: Python, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, LLM Integration (multi-provider tool-calling, prompt engineering), MCP Server Development, NLP/LLMs, Computer Vision

EDUCATION

Master of Science in Computer Science (Machine Learning) 08/2021 – 05/2024

Georgia Institute of Technology, Atlanta, GA

Key Coursework: Machine Learning, Graduate Algorithms, Computer Vision, Computational Photography, AI Ethics

Bachelor of Science in Computer Science (Minor in Mathematics) 08/2014 – 05/2018

Azusa Pacific University, Azusa, CA

PROFESSIONAL EXPERIENCE

Software Engineer 03/2025 – Present

HighPoint Technology Solutions, Inc., Tempe, AZ (Remote)

- Develop and maintain Degree Audit Wizard, a React web application with a Kotlin/GraphQL backend used by universities to configure and preview degree requirements.
- Architected the HighPoint LLM API, a Kotlin library that runs in PeopleSoft with an abstraction supporting 7 AI providers (OpenAI, Anthropic, etc.) with functional parity across all providers.
- Designed the AI Advisor chat widget for the Degree Planner product, featuring LLM-powered conversational guidance with 20+ tool-calling capabilities (path progress summary, GPA simulation, major/minor exploration). Demoed at the March 2026 Vegas EdTech conference, generating significant client interest.
- Led AI developer enablement for all 12 developers across 5 teams: evaluated and integrated AI coding tools (Codex, Claude, Cursor), authored agent context files and prompt libraries, designed a "vibe-coding" workflow with safety guardrails for non-technical staff, and built custom MCP servers for GitLab, Redmine, and Obsidian.

Machine Learning Engineer 11/2024 – 02/2025

Custom Stacks, LLC., Los Angeles, CA

- Engineered a real-time conversational voice AI bot (pipes-voice-ai): implemented audio interruption handling and timing, call closing/farewell flows, and TTS text chunk formatting. Integrated AWS Bedrock (including Nova models), ElevenLabs TTS, and Deepgram STT into a low-latency speech pipeline.
- Built a legal document processing pipeline (manning-law): LLM-powered PDF parsing that extracts structured JSON from legal discovery documents, with OCR fallback using Tesseract (PSM parameter tuning for accuracy), parsing accuracy evaluation, and a Streamlit UI for streaming results.
- Optimized LLM prompts across both projects, reducing API costs, correcting voice mispronunciations, and improving JSON extraction reliability. Initiated observability measures logging inter-service latencies.

Independent Full-Stack Software Developer

01/2023 – 02/2025

Self-employed · Freelance

- Contracted directly with clients to build web applications, clarify requirements, and deliver responsive interfaces and API integrations.
- **Design2Text:** Built a SvelteKit application using the OpenAI API to turn Figma designs into requirements documents.
- **WHIP Cars:** Built an auto-transport load management application with SvelteKit and Supabase under a direct client contract.
- **Shiftwell:** Built a restaurant shift scheduling application with SvelteKit and Firebase.
- **Kiqo:** Developed a web application under a separate direct client contract.

Freelance Full-Stack Web Developer — GeekOffice, LLC

01/2023 – 02/2025

Los Angeles, CA | Freelance engagement

- Partnered with a web designer to build the HyCite corporate website using Angular and Dato CMS.
- Collaborated with the GeekOffice team on smaller web projects, including a showcase website for the company.

Full-Stack Web Developer

05/2017 – 06/2022

Azusa Pacific University, Azusa, CA

- Led front-end development to launch the university web portal, enhancing functionality with PHP, CSS, and JS.
- Collaborated on backend development, integrating databases/APIs and developing secure SQL queries.
- Integrated analytics for strategic insights, managed iOS app (Swift), and reduced API costs by \$180/m (90%).
- Launched a CV occupancy monitoring system and a video streaming service with attendance tracker.
- Mentored junior developers, overhauled the legacy codebase, and established a detailed wiki.

PROJECTS

Open-Source Python Package for ML Engineering

06/2024 – 02/2025

[GitHub](#) | [PyPi.org](#)

- Forked and refactored 10,000+ lines of code in the mlrose-hiive repository for better quality and maintainability.
- Added tests using Pytest, boosting code coverage from 5% to 100%, ensuring dependable functionality.
- Improved documentation with detailed docstrings and modern type hints for better accessibility and usability.
- Fixed critical bugs and optimized performance through NumPy vectorization and algorithm optimizations.

ML Dataset Preprocessing Toolkit

08/2023 – 07/2024

[GitHub](#)

- Created a versatile dataset toolkit, enabling seamless data preparation from various sources (e.g., Huggingface).
- Automated common data preparation tasks like encoding, scaling, normalizing, and sampling data.
- Developed an image-to-tabular conversion pipeline for image-based datasets, utilizing advanced resizing techniques to preserve aspect ratios and enhance downstream ML performance.

Funded Project: Real-Time Campus Parking Lot Occupancy System

01/2018 – 08/2018

Azusa Pacific University

- Collaborated under the mentorship of an AI professor and funding of the IT department to develop a computer vision-based system that monitors campus parking lot occupancy in real-time.
- Created a 3D model of the campus parking lot to generate a synthetic dataset to train a CNN using TensorFlow.
- Deployed the model to the cloud, processing live video feeds to provide real-time occupancy data.